

Prop convergencia uniforme imp medida

Proposición 1. *La convergencia uniforme implica la convergencia en medida.*

Demostración: Sea (X, Σ, μ) un espacio de medida y $\forall n \in \mathbb{N} : f, f_n : X \rightarrow \mathbb{R}$ tales que $f_n \xrightarrow[n \rightarrow \infty]{X} f$. Sea $\lambda > 0$, entonces $\exists N_1 \in \mathbb{N}$ tal que $\forall n \geq N_1 : \|f_n - f\|_\infty < \lambda$. Por definición de $\|\cdot\|_\infty$, $\exists E \in \Sigma : \mu(E) = 0$ y $\forall x \in X \setminus E : |f_n(x) - f(x)| < \lambda$. Tenemos entonces que

$$\begin{aligned} \forall n \geq N_1 : \{x \in X : |f_n(x) - f(x)| > \lambda\} &\subset E \\ \implies \mu(\{x \in X : |f_n(x) - f(x)| > \lambda\}) &\leq \mu(E) = 0. \end{aligned}$$

Por tanto, $\lim_n \mu(\{x \in X : |f_n(x) - f(x)| > \lambda\}) = 0$. ■

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